



2018-Ph-H1-Q1

Section: Electricity

Topic: Current, PD, Power, Resistance

A 12 V power supply is connected across a resistor. The current in the resistor is 2.0 A.
What is the power dissipated in the resistor?

Solution:

Use the equation:

$$P = IV = 12 \times 2.0 = 24 \text{ W}$$

Answer: 24 W

Revision Tip:

Use $P = IV$ when you know both current and voltage. This is the basic power equation for resistive loads.